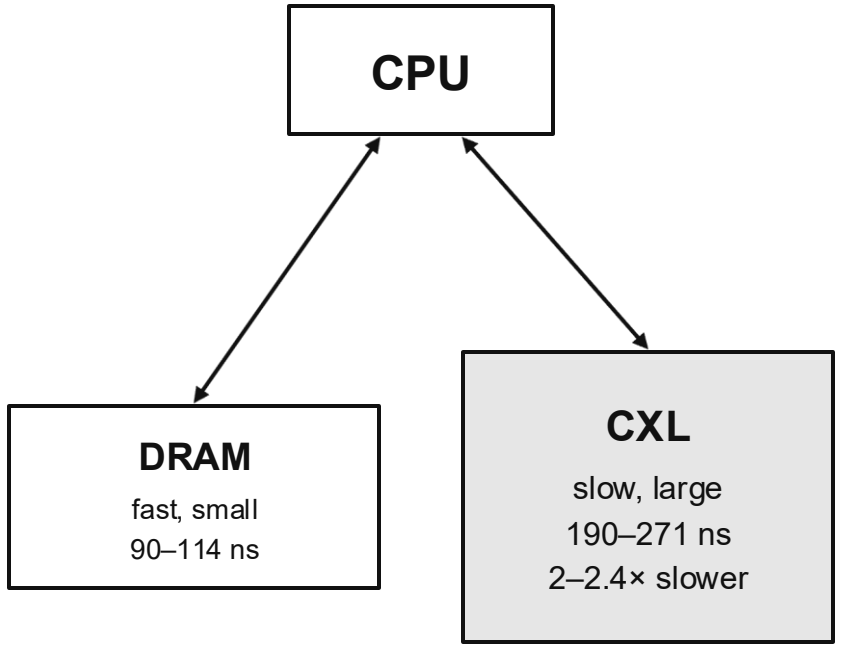
CS-471 Course Project

Tiered Memory Management Beyond Hotness

Team PolyRack-Beyond Hotness: Mayeul Besnard, Sergio Boffi

Problem Statement (1/2)

Why is it important? What is involved?



Which data goes to CXL? Decided by placement and page migration, today based on hotness.

What causes the problem?

Sequential scan: independent misses (high MLP)



→ costs ≈ **1 latency**

Pointer chasing: dependent misses (low MLP)

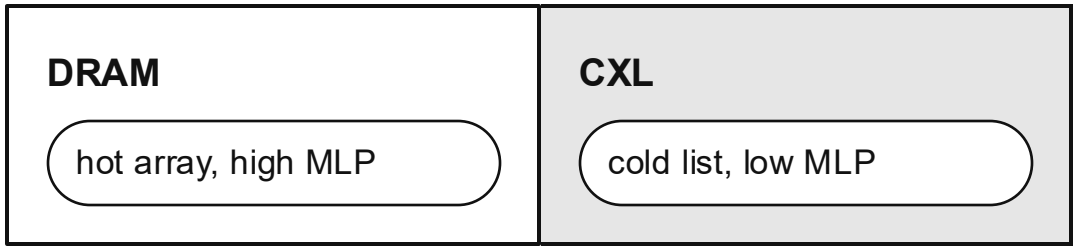


→ costs ≈ **4 latencies**

Effective cost of an access ≈ latency ÷ MLP

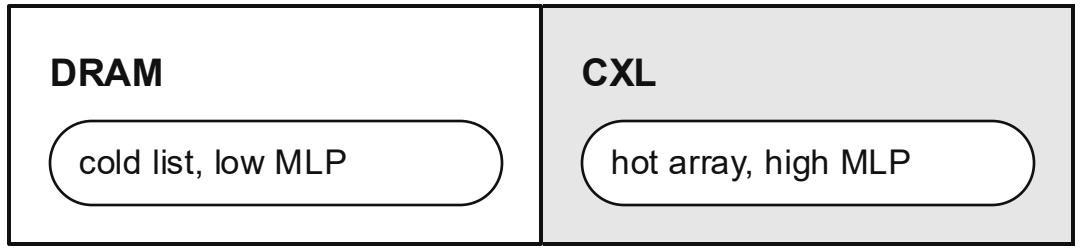
Problem Statement (2/2)

Hotness-based (SOTA)



Slower: the list's serialized misses pay the full CXL latency

Criticality-based (goal)



Faster: MLP helps hidin most of the CXL latency of the array

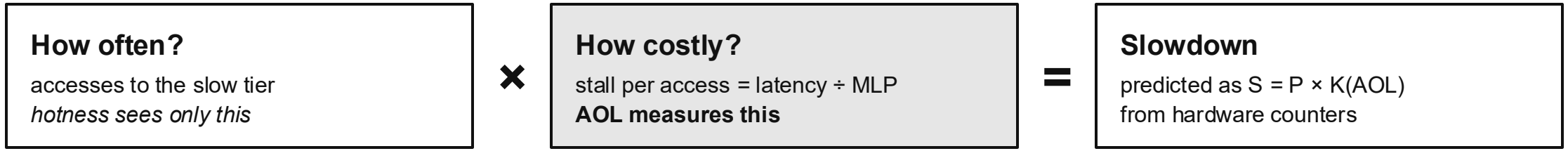
Does prior work solve it? No. TPP, NBT, Nomad and Colloid rank by hotness: 12–14% slower than NoTier (no migrations) on this benchmark, Nomad up to 217% slower on bc-urand.

Questions, Hypotheses and Tests (1/3)

Prioritized questions

- **Q1:** Is hotness a faithful metric for performance? (root)
- **Q2:** If not, can a better metric (AOL) be defined? (only if Q1: no)
- **Q3:** Does AOL improve first placement (SOAR)?
- **Q4:** Do the gains carry over to PolyRack? (only if Q3: yes)

Key insight: why AOL instead of hotness



Questions, Hypotheses and Tests (2/3)

Q1: Is hotness a faithful metric?

- **Hypothesis:** with equal DRAM, Hot-on-DRAM is slower than Cold-on-DRAM
- **Insight:** MLP hides the latency of hot scans
- **Test:** hot/cold microbenchmark, static placements, migrations off

Q2: A better metric?

- **Hypothesis:** P × K(AOL) predicts slowdown better than P and hotness
- **Insight:** stalls per LLC miss is not enough to measure slowdown
- **Test:** measure slowdown and counters, fit K, compare Pearson correlations

Questions, Hypotheses and Tests (2/3)

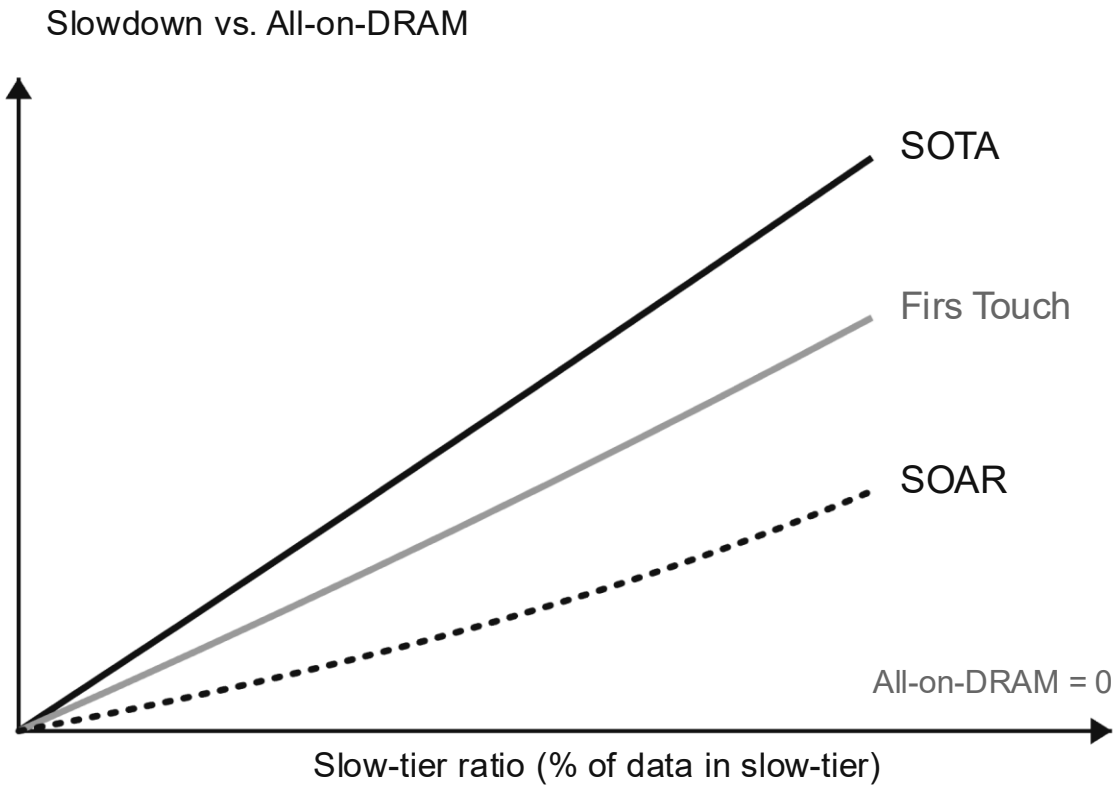
Q3: Better placement (SOAR)?

- **Hypothesis:** SOAR has lower slowdown than NoTier and every SOTA system, same DRAM
- **Insight:** AOL identifies the critical objects
- **Test:** measure slowdown comparing SOTA and SOAR, same workloads, same hardware

Q4: Extends to PolyRack?

- **Hypothesis:** SOAR cut slowdown on PolyRack
- **Insight:** disaggregated memory is also a fast/slow hierarchy
- **Test:** same workloads on PolyRack, only if Q3 holds

Final Deliverable



Concrete goals

- Answer Q1–Q3 on our hardware
- If time permits: Q4 on PolyRack

Concrete results

- X: slow-tier ratio; Y: slowdown vs. All-on-DRAM
- Trend: SOAR lowest slowdown; SOTA can exceed NoTier

Thank you! Open questions for Q&A